
Introduction

Transdisciplinary education requires different learning activities and teaching methods than you might be used to. Learning takes place through collaboration — with society, with other disciplines, and with yourself.

To design this type of education, it's important to understand how the learning process works. You can read more about this under the Theory tab.

On the **Practice page**, you'll find an overview of useful learning activities and teaching methods. Let yourself be inspired and start experimenting.

Theory

Learning in transdisciplinary education is something you do

Students don't just learn *about* complex issues — they work *in* and *on* them together with others. This requires **active, participatory, and creative learning activities**, spread across all phases of the educational project.

This page offers learning activities that align with transdisciplinary learning outcomes **and** with the process of **collaborative integration**.

Overview: Learning Activities by Phase of a Transdisciplinary Project

Phase	Examples of Learning Activities
1. Verkenning van het complexe vraagstuk	- Field visits and context observations - Stakeholdermapping - Storytelling with residents or experts by experience - Visualizations (mindmaps, soft systems)
2. Joint formulation of questions and objectives	• Dialogue sessions with policy-makers • Participatory workshops • Sessions with stakeholders

3. Project setup and team formation	• Feeding up with team roles • Pressure are needed to get together • design sessions (project
4. Execution & integration	• Qualitative quantitative types of interventions, • hybrid disciplinary integrating • Reflective group discussions • Open feedback and perspectives
5. Dissemination & follow-up	• Research academic venues • aimed to practice and

	• Following questions or
Ongoing (in all phases)	• Reflections (and group) • Mapping from (what to) • Relating dialogues

Note: Involve students in all phases — including the start and conclusion of the project. These moments often contain valuable learning experiences, such as building relationships, dealing with uncertainty, or making impact visible.

Other Inspiring Learning Methods

In addition to phase-specific learning activities, you can use methods that stimulate **collaboration, perspective-shifting, and creativity** (Oudenampsen et al., 2024):

Learning activity	Goal
MOOCs / e-modules / field trips	Acquiring knowledge from other disciplines or contexts
Portfolio	Making personal development and role awareness visible
Multidisciplinary debate	Practicing perspective-shifting and position-taking
Disciplinary self-assessments	Raising awareness of one's own frame of reference

Real-life action game	Experiencing how disciplines clash or reinforce each other
Peer feedback	Learning from others' methods and perspectives

Sources of Inspiration

- [Pressure Cookers: School simulations of integrative](#)
- [Development of a Peer Feedback Tool for Design Thinking](#)

References:

Horn, A., Urias, E., & Zweekhorst, M. B. M. (2022). Epistemic stability and epistemic adaptability: interdisciplinary knowledge integration competencies for complex sustainability

Oudenampsen, J., Das, E., Blijlevens, N., & van de Pol, M. H. (2024). The state of the empirical evidence for interdisciplinary learning outcomes in higher education: A systematic review. The Review of Higher Education, 47(4), 467-518.

Practice

Transdisciplinary education requires different learning activities and teaching methods than you might be used to. Learning takes place through collaboration — with society, with other disciplines, and with yourself.

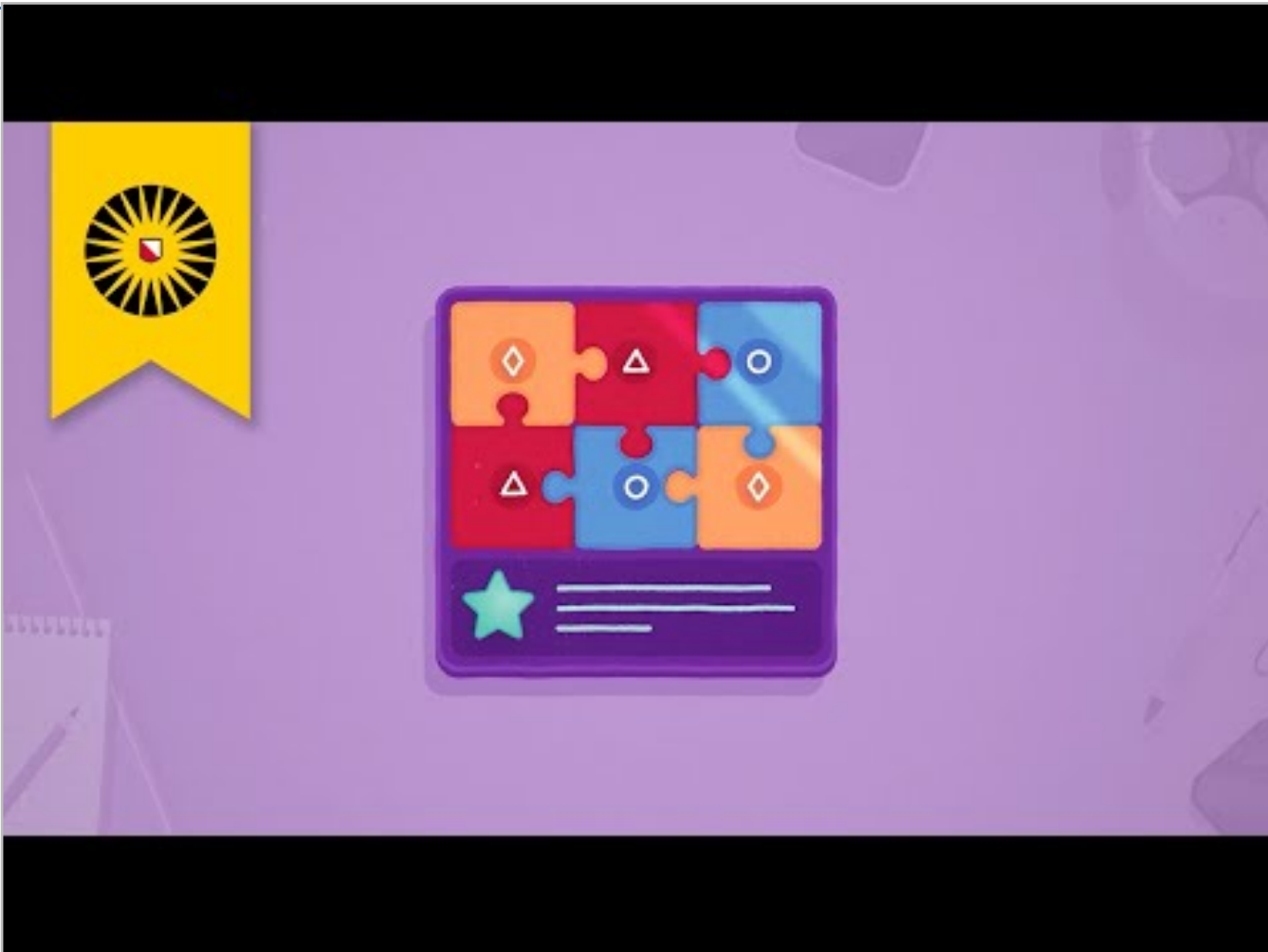
A key feature of transdisciplinary education is the **integration of different perspectives**: scientific, professional, experiential, and societal insights are brought together. This integration process happens through **co-creation** — in collaboration between students and other stakeholders.

This is referred to as **collaborative integration**. Students contribute their own knowledge and backgrounds — for example, from their studies or personal experiences — and work together to develop shared insights, solutions, or products (Oudenampsen et al., 2024).

Five Integrative Educational Methods

(Lambalgen & van der Tuin, 2024)

To consciously guide collaborative integration, you can use **integrative educational methods**. These are teaching strategies that help students connect different perspectives together.



Methode	Focus van integratie

Interdisciplinary Research Process	Disciplinaire insights via common ground
Travelling Concepts	Different conceptual lenses
Design Thinking	Problem- and solution-oriented integration
Object Orientation	Analysis & reintegration of an object
Systems Thinking	System levels and interactions

Each method supports different learning outcomes. Choose what fits your learning objectives, context, and student group.

More information about these methods can be found here: [Chapter 7: Teaching integration in: Handbook of Interdisciplinary Teaching and Administration](#)

References:

Van Lambalgen, R., & van der Tuin, I. (2024). Teaching integration. In: *Handbook of interdisciplinary teaching and administration* (pp. 114-135). Edward Elgar Publishing.

Oudenampsen, J., Das, E., Blijlevens, N., & van de Pol, M. H. (2024). The state of the empirical evidence for interdisciplinary learning outcomes in higher education: A systematic review. *The Review of Higher Education*, 47(4), 467-518.